

Bitcoin Market Sentiment & Hyperliquid Trader Performance Analysis

An end-to-end Machine Learning and Natural Language Processing (NLP) framework built to explore, quantify, and model the hidden relationships between market-wide retail psychology and high-frequency trader performance.

This repository was developed as part of the operational technical assignment for the **Artificial Intelligence (AI) - Internship**.

Key Insights & Project Findings

- **Behavioral Over-Exposure (FOMO):** The Exploratory Data Analysis reveals a strong behavioral bias where retail traders significantly scale up their average position size (Size USD) during market **Extreme Greed** environments, signaling systemic FOMO.
- **The Contrarian Alpha:** The absolute highest net trader win-rates are generated by counter-cyclical execution—specifically executing **BUY** actions against the dominant market trend during periods of panic and **Extreme Fear**.
- **High-Signal Predictive Power:** By blending raw execution metrics with behavioral sentiment vectors, the predictive machine learning model classifies trade outcomes (Wins vs. Losses) with a staggering **82.00% Accuracy** and an **ROC-AUC score of 0.9152**.

Implemented Workflow & Technical Steps

The project was constructed using a modular, pipeline-oriented architecture broken down into four foundational steps:

Step 1: High-Frequency Data Synchronization

- **The Problem:** The Crypto Fear & Greed dataset operates on a daily global rhythm (YYYY-MM-DD), whereas the Hyperliquid trading ledger records massive volumes of high-frequency trades at granular timestamp intervals in Indian Standard Time (DD-MM-YYYY HH:MM).
- **The Implementation:** Built an automated preprocessing pipeline (src/data_preprocessing.py) to parse, clean, and standardize distinct datetime objects from both files. Extracted matching temporal states to join **211,218 raw trades** seamlessly with their respective daily sentiment index variables without data leaks.

Step 2: Exploratory Behavioral Analysis

- Computed hard descriptive statistics tracking total metric volume, collective realized Closed PnL, normalized win rates, and relative exposure size.
- Grouped trading accounts by emotional market regimes to statistically map how execution metrics warp when markets transition from panic (Extreme Fear) to euphoria (Extreme Greed).

Step 3: Predictive Profitability Modeling (Machine Learning)

- Formulated trading success as a binary classification problem: Is_Profitable (1 for realization metrics > 0 , 0 for losses or break-evens).

- Engineered tailored machine learning features including trade_hour, trade_day_of_week, encoded_side, and a custom-calculated contrarian_score (identifying buys during fear thresholds and shorts during greed levels).
- Trained a high-performance **Random Forest Classifier** using a stratified split (stratify=y) to gracefully account for trading outcome imbalances. The model achieved:
 - **Precision (Class 1):** 0.84
 - **Recall (Class 1):** 0.71
 - **Overall Accuracy:** 82%
 - **ROC-AUC Score:** 0.9152

Step 4: Narrative Synthesis via Local Transformer (NLP & Hugging Face)

- **The Implementation:** Implemented a pure, hands-off zero-shot Natural Language Generation pipeline (src/nlp_pipeline.py) using Hugging Face's transformers library running a local text-generation language model framework.
- **The Outcome:** The script automatically references live data structures and metrics directly out of code memory channels, passes them into the neural network architecture, and lets the model autonomously author a coherent, technical summary block reflecting the market findings.